

Shubhankar Gupta

AI Engineer — LLM Applications · RAG & Evaluation · Agent Tooling (MCP) · Python · Production Systems

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github.com/shubhankar360 | linkedin.com/in/shubhankar-gupta-a73aa21a0 | shubhankar360.github.io — portfolio | solquara.com — live production platform I built and run

SUMMARY

AI engineer who builds LLM applications, evaluation harnesses and agent tooling, and runs a production platform end to end. I built and operate the whole platform behind **solquara.com**, an AI video-ad and automation studio — a 40-route site, an accounts and entitlements system covered by **125 end-to-end checks**, and a generative-media pipeline — and published **aistudio-mcp**, an MCP server that lets Claude drive Gemini, Veo and TTS. I publish evaluation work with the numbers: **agentcheck** showed a transcript-only grader agreeing with ground truth worse than chance on a scripted tool-using agent, and my retrieval benchmarks show when hybrid search helps (nDCG@10 0.712 on BEIR SciFact) and when it does not. I profile before optimising and evaluate before claiming quality. Self-taught, after eight years running operations and client service for a family business.

TECHNICAL SKILLS

LLM applications: Anthropic Claude · OpenAI API · Gemini API · LLMs · AI agents (tool / function calling) · structured JSON output · prompt engineering · multimodal generative AI (image, video, TTS)

Retrieval (RAG): RAG · hybrid & semantic search (BM25 + dense embeddings) · reciprocal rank fusion · reranking (lexical; cross-encoder implemented) · embeddings · citation grounding & refusal guards

Evaluation: LLM evaluation (evals) & retrieval evaluation · pass[^]k, Cohen's kappa, McNemar & bootstrap CIs · grader validation against ground truth · Recall@k, MRR, nDCG · pytest / Vitest regression suites in CI

Agent tooling: Model Context Protocol (MCP) · JSON-RPC · long-running jobs (poll & resume)

Backend & data: Python · FastAPI · Pydantic · Streamlit · REST APIs · PostgreSQL + pgvector as a vector database (vector search, HNSW, full-text, BM25) · Docker / Compose · SQL / SQLite / MySQL · Node.js · PHP 8 / WordPress plugins · webhooks (HMAC signing, replay protection) · web crawling & data pipelines

ML: scikit-learn · pandas · NumPy · NLP · classification & regression

Web & delivery: JavaScript (ES2022) · TypeScript · WebGL / GLSL · performance profiling · GitHub Actions CI · Git

Learning now, not yet in production: LangChain / LangGraph · LlamaIndex · Chroma / FAISS

EXPERIENCE

Founder & Lead Engineer — Solquara (solquara.com) Jul 2026 – Present

- AI video-ad & automation studio, remote. Only engineer: architecture, build, AI pipeline, testing, deployment and incident repair on a live production platform.
- Designed the **generative-media pipeline** behind the studio's video ads: keyframe images, then image-to-video, then voiceover. Priced models before committing: on a 25-second Short, keyframe-then-animate cost **~4x less** than Higgsfield's built-in narrated-video workflow for the same length. Built **aistudio-mcp** (below) so Gemini, Veo and TTS generation can be driven from Claude.
 - Built the **accounts, plans and entitlements platform** as a PHP plugin: auth, four plan tiers, monthly usage limits, a one-time free sample keyed on user and canonical email, and billing webhooks with signature checks and replay protection. Wrote **125 end-to-end checks** — IDOR, sessions, login throttling, every subscription state — run in a local WordPress sandbox on real PHP 8.3, and **112 against production** through a key-protected, self-expiring test harness.
 - Shipped an **instant ad-concept writer**: a visitor names their business and trade and reads a storyboarded 10-second ad in about 20 seconds. Chose a deterministic generator over a per-visitor LLM call — zero marginal cost, and the same input always reproduces the same concept, which is what makes every result a shareable link. The concept becomes the brief for the free sample.
 - Rebuilt the public site as **40 routes** (marketing pages, journal, nine browser-side tools, auth pages and a member dashboard) in vanilla ES2022 and WebGL, served ahead of WordPress by a custom plugin. Homepage at launch: **23 stylesheets to 2, 26 scripts to 1, HTML 123 kB to 36 kB**.
 - Profiled a scroll-driven UI and found backdrop-filter taking **~80% of the frame budget** (6 fps to 55 fps once replaced). Replaced device-spec sniffing, which provably excluded capable hardware, with a guard that measures real frame rate and degrades gracefully.
 - Built a **B2B prospecting pipeline in Python**: OpenStreetMap extraction, a resumable website crawler, and Companies House verification that keeps only active limited companies with a role address. Output: **~920 verified prospects**. Also built regional pricing in 10 currencies from one formula shared by build and browser.
 - Caught a failure every automated check missed: a duplicate animation left the **entire pricing table invisible while all HTML-level tests passed**. Made two rules structural — one animation per element, and nothing hidden for a reveal unless JavaScript is running — and the build audit now fails any price not already rendered in the HTML.

Independent ML & AI Practitioner — Self-directed projects 2024 – Present

- Built and published the RAG, evaluation and agent-tooling repositories below. Every AI repo runs on a clean checkout with no API key through offline fallbacks, and CI asserts it on each push.
- Operations & Client Services — Family handicraft & handloom business, Delhi** 2016 – 2024
- Ran daily operations end to end — inventory, fulfilment, pricing and after-sales — with no supervisor. Handled 20+ client conversations a day and used sales and stock data to decide what to buy.

PROJECTS

agentcheck — grading tool-using agents on what they did, not what they said Python · Anthropic SDK · evaluation

- Evaluation harness for tool-using agents: a deterministic back-office world with 7 tools and 36 tasks, graded on **end state**, a zero-tolerance **policy check** over the action log, and the facts in the answer. The environment enforces physics but not policy, so violations are caught rather than hidden.
- On scripted reference agents with known behaviour, measured a transcript-only grader agreeing with ground truth **worse than chance (Cohen's kappa = -0.40)** on an agent that skips checks, passing 16 runs that broke policy, and a 96% pass@1 agent falling to **75% pass[^]8** under repeated trials (unbiased estimators).
- Regression gate pairs an exact McNemar test by task with zero-tolerance policy checks. Claude adapter on the Anthropic SDK with a hand-owned tool loop, tested against a local mock of the Messages API; 22 tests, and CI regenerates the published report.

pg-hybrid-search — hybrid search inside Postgres, measured on BEIR SciFact

TypeScript · PostgreSQL · pgvector · Docker

- Full-text, BM25 and pgvector retrieval fused with reciprocal rank fusion in **one SQL statement**, benchmarked on BEIR SciFact (300 claims, 5,183 abstracts): BM25 + vector hybrid reached **nDCG@10 0.712** vs 0.643 for vectors alone, while hybrid on Postgres's built-in ts_rank_cd made results **worse** (0.597).
- Found that pgvector's default ef_search silently caps LIMIT 100 at **40 rows** (Recall@100 drops from 0.925 to 0.875), and that raising it can make the planner drop the HNSW index on small tables (same results, 10x slower); both documented in the README.
- Same test suite runs on PGLite (Postgres 18 in WASM) and on a real pgvector server in CI, which caught two portability bugs the in-process run could not; Docker Compose stack verified in CI. MiniLM embeddings via onnxruntime-web, parallel and resumable.

aistudio-mcp — Gemini, Veo 3.1 and TTS as an MCP server for Claude

Node.js · MCP · Gemini API

- Zero-dependency Model Context Protocol server exposing nine tools: Gemini text and vision, image generation and editing, Omni Flash video, Veo 3.1 video with native audio, TTS voiceover, the Files API and live model discovery. JSON-RPC over stdio implemented from the spec.
- Long renders return a job id and resume later instead of timing out a conversation. Errors come back with a hint the model can act on (404: re-list models; 429: back off). Media is found by walking the response narrowest-scope first, so an echoed input image is never mistaken for the output.
- **12 end-to-end tests** spawn the real server and point it at a local mock of Google's API, asserting on the exact upstream requests. No key, no network. CI on Node 18, 20 and 22.

hybrid-rag-eval — does hybrid search actually beat BM25?

Python · retrieval evaluation

- On a small, vocabulary-consistent corpus, fusion did not beat its best leg and BM25 won Recall@3 while running **40x faster**. 27 tests; the benchmark itself runs in CI.
- Reproducible benchmark of four retrieval stacks — BM25, dense, hybrid RRF, hybrid + rerank — over 56 passages and 30 labelled queries, scored on Recall@k, MRR, nDCG and latency. BM25 and RRF implemented directly so a result is attributable to a stage; runs offline on LSA embeddings and a lexical reranker, with sentence-transformers and a cross-encoder as drop-in upgrades.

rag-document-qa — cited answers, or an honest refusal

Python · FastAPI · Streamlit · Claude

- RAG service with recursive chunking, vector retrieval and answers that cite the exact passage. Measured that a similarity floor on the offline embedder could not detect unanswerable questions — an off-topic query scored **0.907 vs a relevant one's 0.905** — and replaced it with a vocabulary-coverage guard that separates 8 on-topic from 6 off-topic test queries with a wide gap. **65 tests**, all offline.

job-radar — which remote jobs can a candidate in India actually get?

Python · SQLite · data pipeline

- Pipeline over nine sources (Greenhouse, Ashby and Lever boards for 138 companies, Hacker News, five aggregators): **~12,500 postings a run**, filtered to target roles, then given an eligibility verdict with a reason and an itemised score whose parts are tested to add up; SQLite history flags reposted roles and enforces per-company application caps.
- Evidence-constrained CV tailoring: selects and reorders facts from a profile and never writes new ones, enforced by a test; renders a two-page PDF with headless Chrome. 67 tests, standard library only.

promptdesk — LLM support agent with RAG and escalation

Python · FastAPI · Claude / OpenAI

- Grounds answers in a retrieved knowledge base, classifies sentiment with a chain-of-thought prompt, adapts tone, and uses a structured-output prompt to escalate and open a ticket. Runs in demo mode with no key; 19 tests in CI.

MACHINE LEARNING PROJECTS

- **Heart disease classification** — 13 clinical features, 303 patients; Logistic Regression beat Random Forest and KNN on a held-out split at **88.5% test accuracy**, and the tuned model scored **F1 ≈ 0.87, recall ≈ 0.92** under 5-fold CV.
- **Wine quality prediction** — **13 models** benchmarked as parallel regression and classification problems; Random Forest led both on the held-out split at R² 0.52 and 93% accuracy / 0.95 ROC-AUC.
- **Movie recommender** — content-based recommender over TMDB 5000 with an NLP feature pipeline and cosine similarity.

EDUCATION

Bachelor of Arts, Tourism — Indira Gandhi National Open University (IGNOU)

2020

CBSE Class XII — Air Force Bal Bharati School, New Delhi

2014

Engineering is self-taught, and the evidence is public at github.com/shubhankar360: every project in the Projects section has its own tests and CI.

ADDITIONAL INFORMATION

Languages: English (fluent, professional) · Hindi (native)

Remote: dedicated home office, reliable connectivity, async-first with written decisions.